

obj-025 T7: SA as a Proxy for Variational Free Energy Minimization

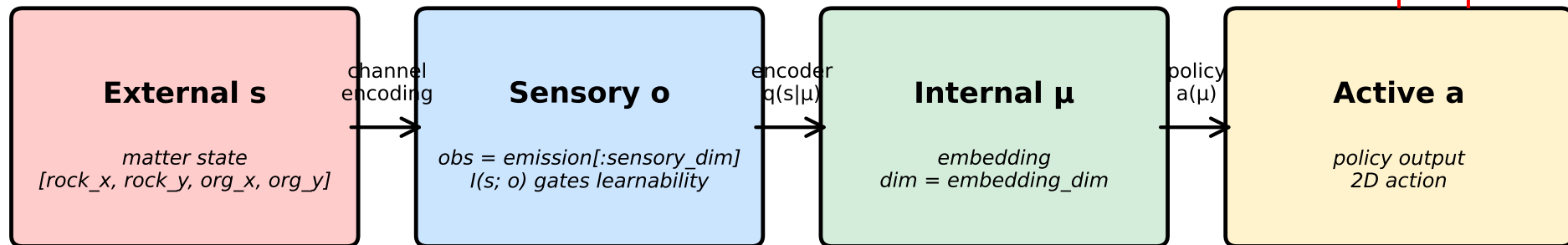
SA $\equiv \mathbb{E}_s[\langle a(\mu(o(s))), a^\circ(s) \rangle]$ — a bounded estimator of $-\text{KL}[q(s|\mu) \parallel p(s|o)]$

Oracle $a^\circ(s)$

optimal action given full state

$F = D_{\text{KL}}[q(s|\mu) \parallel p(s|o)] - \ln p(o) \rightarrow$ minimized by matching μ to posterior

cosine
similarity



environment: action perturbs state

Markov blanket partition: $\{o, a\}$ separates internal μ from external s